

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

A neural network is a machine learning model that learns patterns from training data and uses those patterns to make predictions on new data. In this case, the neural network is used to predict student scores. The model achieved a **Training Mean Squared Error (MSE) of 0.85** and a **Testing Mean Squared Error (MSE) of 5.20**.

Mean Squared Error (MSE) is a performance metric used to measure the average squared difference between the predicted values and the actual values. A lower MSE indicates better prediction accuracy. Since the training MSE is very low (0.85) while the testing MSE is much higher (5.20), it indicates that the model performs well on the training data but poorly on unseen data. This situation is called **overfitting**.

Reasons for the Difference

1. Overfitting

The most common reason for this difference is overfitting. The neural network has learned the training data too well, including its noise and minor details, instead of learning the general patterns. As a result, it cannot make accurate predictions on new test data.

2. Small Training Dataset

If the training dataset is too small, the model may not be exposed to enough examples to learn general patterns. Instead, it memorizes the available data, leading to poor performance during testing.

3. Complex Neural Network Architecture

A neural network with too many hidden layers or neurons has a very high learning capacity. Such a model can memorize the training data rather than generalize from it, causing a significant increase in testing error.

4. Presence of Noise and Outliers

If the training data contains incorrect values, missing data, or outliers, the model may learn these unwanted patterns. This reduces its ability to predict accurately on clean test data.

5. Excessive Training (Too Many Epochs)

Training the neural network for too many epochs can make it fit the training data perfectly. Although the training error decreases, the model loses its ability to generalize to new data.

6. Lack of Regularization

Regularization techniques such as Dropout, L1 Regularization, and L2 Regularization help prevent overfitting. If these techniques are not used, the neural network may become overly dependent on the training data.

7. Poor Data Distribution

If the training data and testing data are not from the same distribution, the model may perform well during training but fail during testing because it encounters different patterns.

Corrective Measures

1. Increase the Size of the Training Dataset

Providing more training examples helps the neural network learn general relationships instead of memorizing individual samples. More data usually improves the model's ability to generalize.

2. Apply Regularization Techniques

Regularization methods help reduce overfitting.

- **Dropout:** Randomly disables some neurons during training.
- **L1 Regularization:** Reduces unnecessary model weights.
- **L2 Regularization:** Prevents weights from becoming too large.

These techniques improve the model's performance on unseen data.

3. Use Early Stopping

Early stopping monitors the validation error during training. When the validation error starts increasing while the training error continues decreasing, training is stopped. This prevents the model from overfitting.

4. Simplify the Neural Network

Reducing the number of hidden layers or neurons decreases the complexity of the model. A simpler model is less likely to memorize the training data and is more likely to generalize well.

5. Perform Cross-Validation

Cross-validation divides the dataset into multiple training and validation sets. This provides a more reliable estimate of model performance and helps in selecting the best model.

6. Improve Data Preprocessing

Data should be cleaned before training by:

- Removing outliers.
- Handling missing values.
- Normalizing or standardizing features.
- Removing irrelevant features.

Better-quality data leads to better predictions.

7. Tune Hyperparameters

Hyperparameters such as learning rate, batch size, number of epochs, and optimizer should be adjusted carefully. Proper tuning improves both training and testing performance.

Conclusion

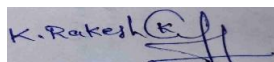
The neural network achieved a **Training MSE of 0.85** and a **Testing MSE of 5.20**, indicating a large gap between training and testing performance. This is a clear sign of **overfitting**, where the model memorizes the training data instead of learning general patterns. The main reasons include a complex model, limited training data, excessive training, lack of regularization, noisy data, and poor data distribution.

To overcome this problem, techniques such as **increasing the dataset size, applying regularization, using early stopping, simplifying the network architecture, performing cross-validation, improving data preprocessing, and tuning hyperparameters** should be used. These corrective measures help the neural network generalize better and improve its prediction accuracy on unseen data, resulting in lower testing error and a more reliable model.

Code of Conduct:

I Kothakota Rakesh/192525075 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided